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RESEARCH ARTICLE

CowNet-AI: A Multi-Agent Decision Support Framework for Social Network–Driven Welfare Insights in Dairy Cattle

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ABSTRACT Dairy farms increasingly deploy sensor networks that generate high-frequency behavioral and spatial data, yet translating these multimodal streams into actionable welfare insights remains a key challenge. This study presents CowNet-AI, a modular, multi-agent conversational decision-support framework that integrates social network analysis with large language model–driven agents to deliver explainable welfare risk indicators and simulation-based insights. The system comprises a behavioral simulation engine for synthetic herd interaction generation, a data ingestion pipeline, social network analysis modules computing centrality- and isolation-based metrics, and a LangGraph-based multi-agent architecture including supervisor, data loader, SNA, research, simulation, response, and reporting agents. The framework is evaluated using the MmCows dataset, with demonstrations conducted through natural-language farmer queries. Experimental results show that CowNet-AI achieves high intent routing accuracy and a low hallucination rate (5%), outperforming baseline single-agent LLM and static rule-based SNA approaches. The system effectively integrates behavioral metrics with contextual knowledge to generate personalized, explainable recommendations, while enabling autonomous query routing and scenario-based reasoning. End-to-end latency varies depending on agent configuration and query complexity. This work represents a pilot-scale validation conducted on a limited dataset, establishing the feasibility of agentic, explainable AI systems for livestock welfare intelligence. Current limitations include reliance on behavioral proxies without direct clinical validation, dependence on LLM-based evaluation, and computational scaling challenges for larger herds. Future work will focus on multimodal data integration, expert-in-the-loop validation, and system optimization for real-time deployment.

INDEX TERMS Conversational AI, dairy cattle welfare indicator, digital livestock management, explainable AI, large language models, multi-agent systems, precision agriculture, simulation-based reasoning, social network analysis.

I. INTRODUCTION

The global dairy sector is rapidly adopting sensor-based technologies to monitor herd behavior and health [1], [2]. Ultra-wideband localization systems, computer vision pipelines, activity monitors, and barn-level environmental sensors now generate large volumes of spatial, temporal, and behavioral data each day [3], [4], [5]. This instrumentation offers unprecedented visibility into herd dynamics, but the central challenge remains: how to convert raw data into reliable, farmer-actionable management decisions.

In current practice, detection of implicit or explicit welfare issues such as social isolation, aggressive encounters, group fragmentation, and overcrowding relies heavily on the subjective judgement of farm managers and veterinarians [6]. These assessments are labor-intensive, inconsistent across

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observers, and largely reactive rather than preventive. Recent work in animal welfare science has shown that social network analysis, as a graph-based approach to relational structure, can quantify isolation, centrality, and dominance in cattle herds in a more objective way [7]. Existing implementations, however, typically produce static reports that are difficult to integrate into day-to-day management and are not coupled to interactive decision-support or explainability mechanisms.

In parallel, the emergence of large language models and multi-agent orchestration frameworks such as LangGraph [8] has opened opportunities for intelligent systems that blend quantitative analysis and natural-language interaction [9]. Multi-agent LLM systems can coordinate specialized tools, maintain memory, and provide stepwise reasoning in response to user queries. To date, these capabilities have not been operationalized for real-time herd welfare risk criteria or integrated with social network analysis in an agricultural setting.

This paper introduces CowNet-AI, an end-to-end decision-support system that addresses these gaps. CowNet-AI combines a behavioral simulation engine for synthetic data generation and stress testing, a schema-aware data pipeline, social network analysis modules for graph-based metrics, and a LangGraph multi-agent architecture that supports open-ended farmer queries. Explainability mechanisms ground recommendations in both computed metrics and relevant research literature.

The main contributions of this work are as follows. First, we present what is, to our knowledge, the first agentic social network analysis-based decision support system that proactively flags livestock welfare risks. We show how a multi-agent LLM architecture can coordinate specialized analytical and computational tasks for herd management. Second, we design the system to provide explainable decision support: recommendations are tied explicitly to underlying metrics and literature, rather than emerging from a black box. Third, we demonstrate open-ended query handling that goes beyond fixed intent catalogues, using a supervisor agent and research agent to interpret and ground natural, farmer-phrased questions. Fourth, we adopt a modular architecture that is extensible to new data sources, analytic modules, and intervention strategies.

The remainder of this paper summarizes some recent relevant research works, describes the system architecture and methods, reports preliminary validation and demonstration results, and outlines a roadmap for quantitative evaluation and deployment-ready optimization.

II. RELATED WORKS

A. SOCIAL NETWORK ANALYSIS IN LIVESTOCK

Social network analysis has recently been applied to livestock to better understand herd dynamics. Grosfilley et al. [10] used weighted networks constructed from proximity data to show that degree centrality, the number of direct social contacts, correlates with disease transmission risk. Marina et al. [11] examined betweenness centrality to identify cows that act as bridges between otherwise separated subgroups, and showed that monitoring such animals can stabilize herd structure. These analyses were extended in [12] to community structure and found that the stability of community membership predicts longer-term welfare outcomes. In all of these studies, social network analysis functions primarily as a post-hoc analytical tool that produces descriptive statistics and visualizations. None of these systems is designed as an interactive decision-support tool, and none is integrated with conversational AI or formal explainability methods.

Recent work shows that social disruption and reduced interaction in cattle are associated with measurable stress responses, production changes, and disease onset, while illness itself manifests as altered social behaviour and reduced connectivity within the herd [13], [14], [15], [16]. These findings support the use of interaction-derived metrics as behaviour-based indicators of welfare-relevant risk rather than direct clinical measures.

B. AI AND DIGITAL LIVESTOCK SYSTEMS

A growing body of work investigates AI in precision livestock farming. Neethirajan [17] reviewed applications ranging from object detection and activity recognition to anomaly detection in dairy and other species, and highlighted that most tools target narrow tasks such as lameness detection or estrus monitoring. Several recent high performance deep learning-based cow detection and tracking models were compared on variations of YOLO and Detectron-2 for visual dairy cattle data in [18]. While these approaches are valuable, they do not reason about herd-level social structure or generate management recommendations. For example, Chen et al. [19] proposed a deep learning framework for behavior classification from video and reported high accuracy on predefined activity labels, but the system does not infer social interactions or identifies animal welfare risks. Similarly, a wearable sensor platform for individual health monitoring was presented in [20], but they explicitly noted that herd-level social dynamics lie outside its scope.

C. MULTI-AGENT LLM SYSTEMS

Multi-agent orchestration frameworks such as LangGraph, built on LangChain, have become important tools for building complex LLM applications. Li et al. [21] showed that agentic systems with tool use and explicit reasoning modules outperform single-pass LLM calls on multi-step reasoning tasks. Wei et al. [22] demonstrated that chain-of-thought prompting improves both accuracy and user trust. Most published demonstrations, however, focus on general-purpose tasks such as code generation or abstract problem solving. Agricultural applications are rare, and we are not aware of any prior work that uses multi-agent LLM systems for animal welfare risk warning. CowNet-AI applies this paradigm in a domain where both transparency and safety are critical.

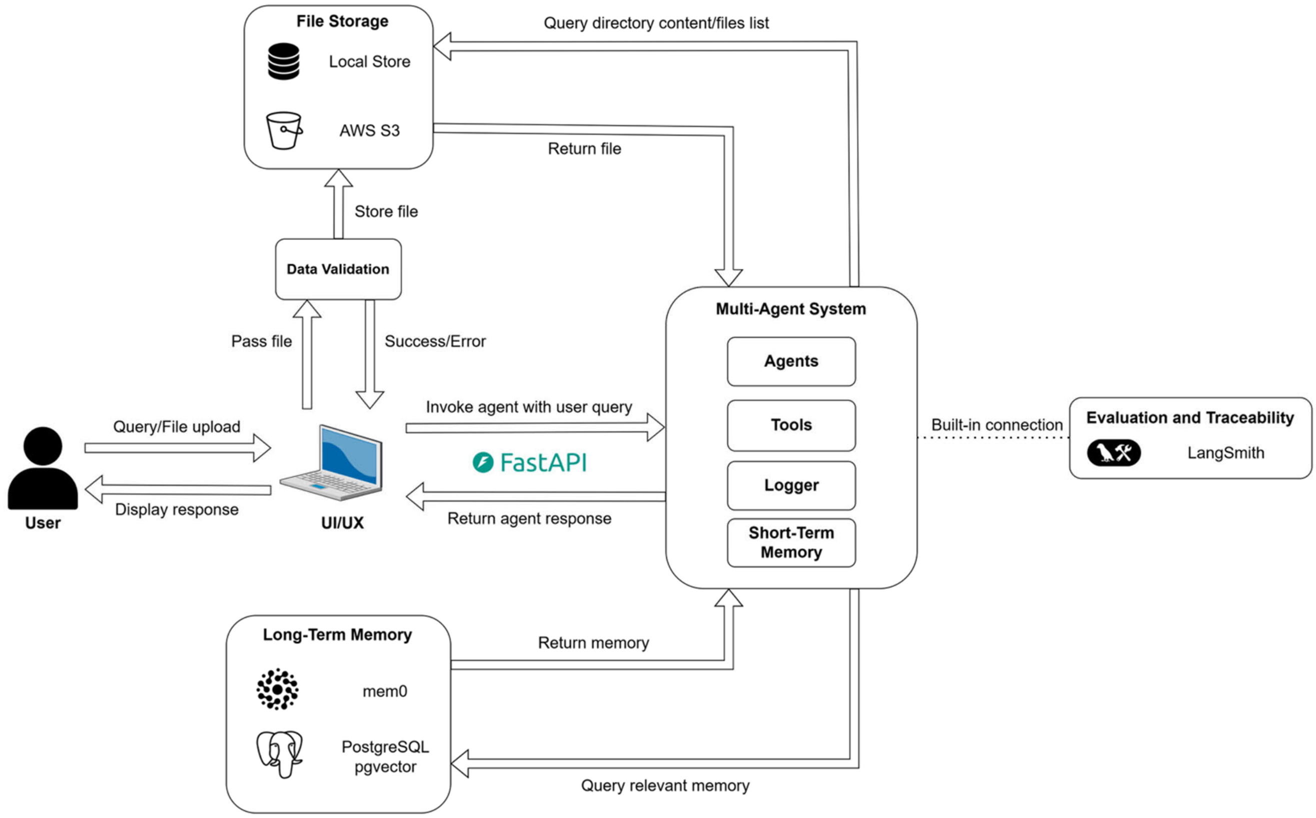

FIGURE 1. End-to-end architecture of the CowNet-AI multi-agent decision-support system showing user interface, API layer, supervisor-controlled agents, memory components, and file storage in a reference deployment.

D. EXPLAINABLE AI AND TRANSPARENCY

Explainability is increasingly recognized as a requirement in AI-based decision systems, especially in safety-critical domains. The American Dairy Science Association [23] has emphasized that farmers must be able to understand why an AI system recommends a particular action in order to incorporate its guidance into management practice. Ribeiro et al. [24] introduced model-agnostic local explanation techniques that have influenced many subsequent designs. In CowNet-AI, we focus on domain-specific explainability: the response agent ties recommendations directly to computed metrics, trends, and cited studies, rather than relying on generic feature importance summaries.

III. SYSTEM ARCHITECTURE OVERVIEW

CowNet-AI is organized into four tightly coupled subsystems (Figure 1). A data ingestion and storage pipeline, built around FastAPI and Python, performs file detection, schema validation, cleaning, and loading into a local or cloud file store. A social network analysis module transforms weekly interaction data into graphs and computes centrality metrics, isolation scores, community structure, and temporal changes using NetworkX [25] and associated scientific Python libraries. A rules engine and LLM reasoning layer translate these metrics into management-relevant intents and support open-ended query handling. Finally, a multi-agent system implemented in LangGraph orchestrates the supervisor, data loader, SNA, research, simulation, response, and report agents to answer farmer queries in a conversational manner.

All agents interact through a shared graph-state memory. This state holds interaction tables, derived metrics, simulation outputs, research summaries, and conversation history. The stateful architecture allows agents to collaborate without direct coupling, to maintain context across multi-turn dialogues, and to support reproducibility and traceability with built-in connection to the LangSmith platform [26]. The architectural separation in CowNet-AI mirrors global recommendations for agent design, which emphasize distinct application, orchestration, and reasoning layers [27].

IV. METHODS AND TECHNICAL IMPLEMENTATION

A. BEHAVIOURAL SIMULATION ENGINE

To test social network analysis (SNA) algorithms independently of scarce or proprietary real farm datasets, a discrete-time simulation models dairy cow behavior in a reference freestall barn (Figure 2) with 20 autonomous cow agents. Each agent has fixed attributes including unique ID, parity (number of calvings), lactation stage (lactating or dry, determining initial pen assignment), home pen ID, and behavioral

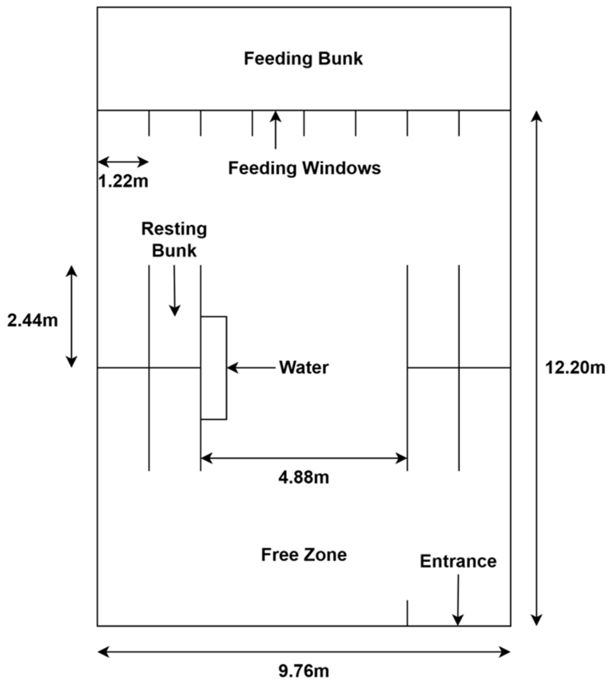

FIGURE 2. Layout of the experimental dairy pen showing feeding, resting, water, and free movement zones used to derive social interaction networks. The pen assignments for the cows, their interactions, and their registry data are used by the agents to compute the social network and provide appropriate responses to user queries.

traits such as friendliness, aggressiveness, and activity level; dynamic states track current activity (resting, feeding, drinking, moving, socializing, idle, or sick), activity timer, health status, spatial position (x, y coordinates), target zone, and pairwise friendship scores that initialize randomly, decay over time, and increment upon interactions.

The barn comprises three lactating pens and one dry pen, each a rectangular grid (Figure 2) subdivided into zones: feeding windows (limited capacity, causing competition), resting bunks (cow-assigned), water troughs, entrances/exits, and free movement areas, with dimensions following freestall specifications from Sardzadeh et al. [29]. Cows transition activities via a probabilistic state machine weighted by traits and needs (e.g., active cows favor movement/socializing, friendly cows seek nearby friends); durations draw from realistic ranges (e.g., resting hours, feeding/drinking minutes). Pathfinding employs the A$*$ algorithm to compute obstacle-avoiding routes around barriers like walls and bunks.

Proximity interactions trigger when cows in the same pen remain within 150 cm for $\geq$60 consecutive seconds, logging cow IDs, timestamps, zone, and duration per Chopra et al. [28]; friendly cows form longer interactions. Sickness occurs probabilistically per second, isolating affected cows to a virtual external pen (excluded from interactions) until random recovery (days). Feeding halts other activities at fixed daily times, with cows queuing at windows; weekly (604,800 s) pen logs track assignments.

Over configurable multi-week horizons at 1-s resolution, the engine outputs three CSVs: interactions (proximity events), cow registry (metadata), and pen assignments, directly feeding SNA and rules modules; emergent patterns match freestall observations.

B. DATA INGESTION AND ORCHESTRATION PIPELINE

CowNet-AI employs a modular data pipeline to securely ingest and validate CSV files containing cow location data, registries, and pen assignments from farm sensors, ensuring only high-quality data reaches analysis modules (Figure 3). This design supports precision livestock farming by handling streaming uploads, automated validation, and flexible storage, minimizing errors in downstream social network analysis.

Users upload files via a web interface button, where the frontend streams them as HTTP multipart requests to a FastAPI backend and writes to temporary storage - local directories or cloud buckets like S3. A background service then validates files by confirming CSV format via content signatures, loading with data processing libraries to check schemas (required columns, types like numeric IDs/datetimes, row limits, missing value thresholds), rejecting invalids with reasons and deleting them [30].

Validated files promote to permanent storage (e.g., /data/validated/ or main cloud buckets). Downstream agents query only validated files, enabling reliable multi-agent processing; this schema-enforced approach aligns with layered architectures for sensor data in dairy digital twins, supporting scalable farm management.

C. SOCIAL NETWORK ANALYSIS MODULE

Weekly interaction logs aggregate into undirected graphs using NetworkX, where nodes represent individual cows and edges denote interactions weighted by count or cumulative duration. Degree centrality quantifies direct connections, indicating social activity and disease exposure risk; betweenness centrality measures control over shortest paths between cows, identifying structural bridges; eigenvector centrality assesses influence by prioritizing connections to other central nodes [11].

Community structure emerges via the Louvain modularity optimization algorithm, which greedily maximizes modularity to partition networks into cohesive subgroups, revealing stable social clusters relevant to dairy welfare [10]. Isolation risk derives as the inverse of closeness centrality; low closeness (high inverse) signals peripheral positioning and welfare vulnerability, as isolated cows exhibit reduced reachability and group integration. The isolation risk score (1) for each cow is computed from the social network metrics and weights based on logical assumptions on their contributions discussed in literatures [31], [32], [33], and [34]. The scores were computed from the weekly social networks and the changes between two consecutive weeks. The degree deficit

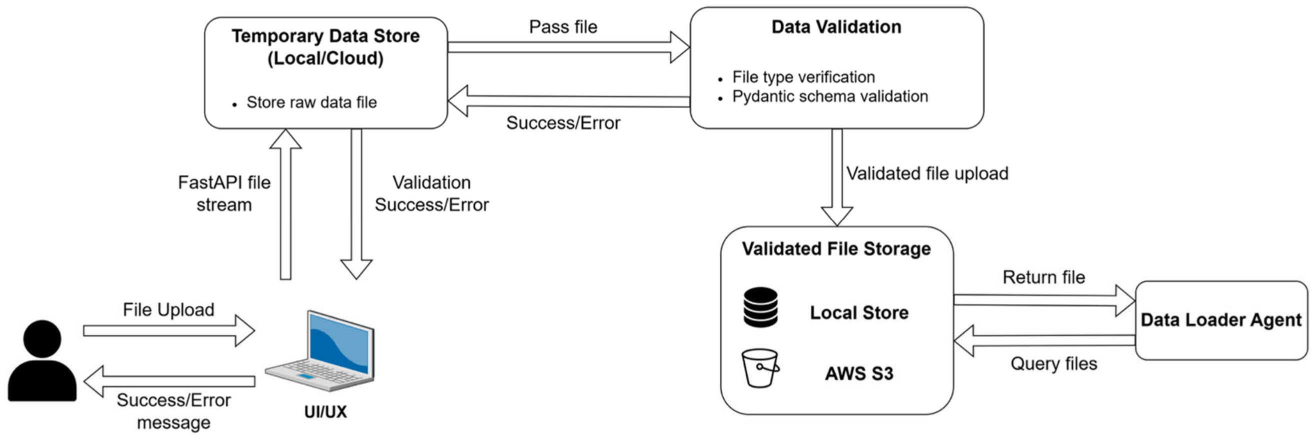

FIGURE 3. Data ingestion pipeline illustrating file upload, schema validation, temporary storage, and retrieval by the Data Loader Agent in a reference local or cloud deployment.

of a cow refers to the differences in the total number of interactions between week 1 and week 2. Centrality decline measures the change in the importance/contribution of the cow in two consecutive weeks. The cow's reachability change between two weeks is defined by the closeness deficit. The frequency deviation of a cow measures the changes in the interaction rate between week 1 and week 2. All scores are normalized to compute the isolation risk for each cow based on the scores calculated from the social networks of two consecutive weeks.

$$\begin{aligned} \textit{Isolation Risk} &= (0.35 \times \textit{Degree Deficit}) \\ &\quad + (0.30 \times \textit{Centrality Decline}) \\ &\quad + (0.20 \times \textit{Closeness Deficit}) \\ &\quad + (0.15 \times \textit{Frequency Deviation}) \end{aligned} \tag{1}$$

D. MULTI-AGENT ARCHITECTURE

The core of CowNet-AI is a multi-agent orchestration layer built with LangGraph, enabling structured collaboration among specialized agents for dairy cow social network analysis (SNA). The agents use structured message passing-based communication within the LangGraph-based framework. All agents share access to a common state dictionary, GraphState, a collaborative data structure containing conversation messages, raw interaction data derived from cow locations, SNA metrics (degree centrality, betweenness, eigenvector, isolation risk, and community disruption) on both a herd level and cow level, simulation results, research summaries. When invoked, each agent can perform specialized tasks using their own set of tools, based on the data and conversation context, as a modular functional unit. Agents communicate indirectly through this shared state, with execution traces captured via LangSmith for debugging and optimization. Our design aligns with emerging best practices for interoperable multi-agent systems that rely on standardized coordination protocols [29].

1) SUPERVISOR ROUTING AND INTENT RESOLUTION

The supervisor agent acts as an LLM-powered router that receives farmer queries, inspects the GraphState (conversation history + analysis artifacts), and infers intent via natural language reasoning (e.g., isolation detection, centrality analysis, simulation scenarios, research-backed recommendations). It selects the next specialist agent and records its rationale in the message history, forming a closed-loop control system that iterates until a complete response is generated (Algorithm 1) [35]. This dynamic routing outperforms static keyword matching, adapting to multi-turn context and missing prerequisites (e.g., routing to SNA agent if metrics are absent). The decision-making process is centralized and maintained by the supervisor agent.

2) SPECIALIST AGENTS

- **Data Loader Agent**: Retrieves cow location data and extracts pairwise cow interactions data populating GraphState with raw edges for SNA computation.
- **SNA Agent**: Constructs NetworkX graphs and computes herd-level (density, diameter) and cow-level metrics (degree centrality, betweenness, eigenvector, isolation risk, community disruption scores), integrating dairy welfare indicators.
- **Simulation Agent**: Manipulates graph structure for "what-if" scenarios by removing a node and recomputes metrics to assess stability impacts.
- **Research Agent**: Queries external APIs (Semantic Scholar, arXiv, PubMed) and the rules/intents knowledge base to provide evidence-based explanations, injecting summaries into GraphState research [36].
- **Response Agent**: Synthesizes SNA metrics, simulations, and research into farmer-friendly explanations. Retrieves long-term context before responding; signals need_more if prerequisites are missing, returning control to the supervisor.

- **Report Agent**: Assembles comprehensive PDF briefings from GraphState artifacts, triggered explicitly by "report" intents.

Figure 4 shows the connections between the agents. Directed conditional edges represent routing decisions based on intent classification. The agents are sequential and they are invoked one at a time by the supervisor's routing. Agents are implemented as LangChain tools with domain-specific prompts and can be powered by any large language model of choice. Execution traces are captured via LangSmith to support debugging and performance optimization.

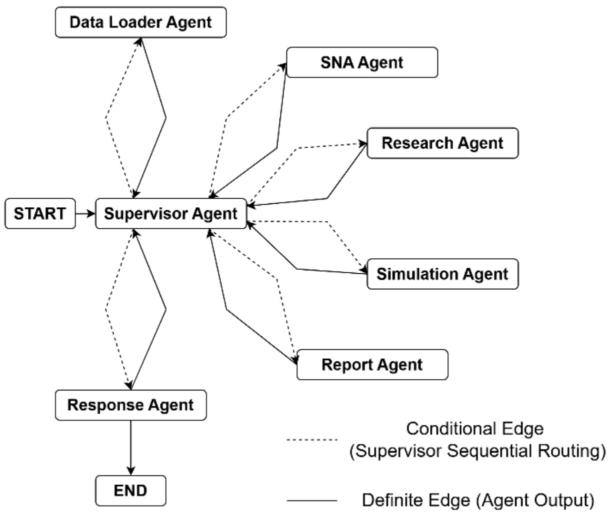

FIGURE 4. Supervisor-controlled multi-agent routing graph. Dashed edges denote conditional routing decisions and solid edges represent agent outputs.

E. RULES ENGINE AND FIXED INTENTS

In this pilot study, CowNet is designed to address user queries that are relevant to the dairy farm data provided by the user and the SNA of the cows. All user queries are accepted by the system without generating any errors, but they are evaluated by the supervisor agent to determine their relevance based on the available farm data and the constructed social networks. Open ended, random or semantically irrelevant queries are filtered out by the system, and no output is generated in such cases. The system simply asks the user for the next query. This design choice ensures that CowNet focuses on meaningful, domain specific analysis while preventing responses to out-of-scope or unrelated open-ended questions.

Alongside open-ended query handling, CowNet-AI maintains a library of fixed management intents (Table 1) that correspond to recurring questions, such as identifying the top isolation-risk cows in a pen, detecting cows whose isolation scores have increased relative to the previous week, simulating the effect of removing a specific cow, and generating weekly PDF summary documents.

Algorithm 1 CowNet-AI Supervisor–Agent Routing and Memory Cycle

Require: q_{user} // new farmer query (text)
Require: S // initial GraphState
Require: M // long-term memory layer
Ensure: S_{final} // final GraphState with response appended
Ensure: y_{final} // final natural-language response
  S.messages $\leftarrow S$.messages $\cup \{q_{user}\}$
  $y_{final} \leftarrow \emptyset$
  done $\leftarrow$ FALSE
  while done = FALSE **do**
    a_{next}, *rationale* $\leftarrow$ SupervisorLLM (S)
    S.messages $\leftarrow S$.messages $\cup \{rationale\}$
    if a_{next} = data_loader_agent **then**
      $S \leftarrow$ DataLoaderAgent(S)
      continue
    else if a_{next} = sna_agent **then**
      $S \leftarrow$ SNAAgent(S)
      continue
    else if a_{next} = simulation_agent **then**
      $S \leftarrow$ SimulationAgent(S)
      continue
    else if a_{next} = research_agent **then**
      $S \leftarrow$ ResearchAgent(S)
      continue
    else if a_{next} = report_agent **then**
      $S \leftarrow$ ReportAgent(S, M)
      continue
    else if a_{next} = response_agent **then**
      $C_{mem} \leftarrow$ RetrieveFromMemory}(M, S)
      need_more, y_{cand}, ΔS, $\Delta M \leftarrow$ ResponseAgent(S, C_{mem})
      if *need_more* = TRUE **then**
        $S \leftarrow$ MergeState($S, \Delta S$)
        $M \leftarrow$ MergeMemory($M, \Delta M$)
        S.messages $\leftarrow S$.messages $\cup$ {"ResponseAgent: needs more info"}
    continue // loop continues, supervisor re-invoked
      else
        $y_{final} \leftarrow y_{cand}$
        $S \leftarrow$ MergeState($S, \Delta S$)
        $M \leftarrow$ MergeMemory($M, \Delta M$)
        $M \leftarrow$ StoreInteraction($M, q_{user}, y_{final}, S$)
        S.messages $\leftarrow S$.messages $\cup \{y_{final}\}$
        done $\leftarrow$ TRUE
      end if
    else
      S.messages $\leftarrow S$.messages $\cup$ {"Unknown agent; aborting"}
      done $\leftarrow$ TRUE
    end if
  end while
  return $S_{final} \leftarrow S, y_{final}$

F. MEMORY PIPELINE

CowNet-AI implements a dual-layer memory pipeline that integrates short-term conversational persistence with long-term semantic memory, enabling stateful, personalized dairy welfare risk analysis across multi-turn interactions (Figure 5). This hybrid approach addresses key limitations in stateless LLM agents - context loss and lack of personalization - while ensuring efficient agent coordination through a shared GraphState.

Short-term memory captures the complete working state of a single conversation thread, including messages, routing decisions, and analysis artifacts (SNA metrics,

TABLE 1. Summary of fixed management intents: Short description of each of the 15 intents, required metrics, agents involved, and output. additionally, all intents include the supervisor agent.

Intent	Description	Required Metrics	Involved Component	Output
Cows more isolated vs last week	List cows whose average distance from herd increased week-over-week	-Weekly isolation scores	-SNA -Response	List of affected cows
Communities and dominant hubs	Detect communities and hubs and flag dominant hubs	-Community membership (Louvain, clique detection) -Centrality scores	-SNA -Response	Community map with hubs highlighted
Simulate moving cow out of pen	Simulate effects of moving a cow out of a pen	-Degree centrality -Betweenness centrality -Eigenvector centrality	-SNA -Simulation	Before-after comparison
Pen assignment recommendations	Recommend pen assignments honoring capacity and protected cow rules	-Pen capacity -Pen occupancy -Protected cow status	-SNA -Response	List of recommendations
Explain cow flag rationale	Provide detailed explanation of metrics and rules triggering a flag	-Isolation scores	-Response -Research	Plain-language explanation
Week summary	Generate one-page summary of key metrics, events, and recommendations	-Weekly SNA metrics -Risk highlights -Major changes	-SNA -Research -Simulation -Report	PDF summary report
Top 5 most central cows	List cows with highest centrality and describe influence	-Degree centrality -Betweenness centrality -Eigenvector centrality	-SNA -Response	Ranked list with roles
New bridges between communities	Identify new bridge cows between communities vs previous week	-Betweenness centrality	-SNA -Response	List of bridge cows
Isolates at lactation risk	Flag isolated cows at risk due to lactation stage	-Isolation score -Lactation stage -Risk threshold	-SNA -Response	Risk-flagged list
Pens near capacity	Flag pens approaching or exceeding safe capacity	-Pen capacity -Occupancy -Density threshold	-SNA -Response	Warning list with suggestions
Major changes delta report	Summarize week-over-week changes in metrics and structure	-Delta in centrality metrics -Isolation scores -Pen assignments	-SNA -Response	Delta report highlights
Splitting community "what-if"s	Simulate effects of splitting a community on metrics	-Social network graph -Degree centrality -Betweenness centrality -Eigenvector centrality	-Simulation -Response	Predicted risk/benefit comparison
Centrality drops to monitor	Flag cows with significant week-over-week centrality decrease	-Delta in centrality metrics -Connectedness -Isolation scores	-SNA -Response	Monitoring list
Export recommendations checklist	Compile all current recommendations into actionable checklist	-All active recommendations	-Report	PDF checklist
Log farmer decisions	Record farmer feedback and rationale for recommendations	-Recommendation ID -Farmer response -Rationale	-Long-Term Memory	Stored feedback log

simulation graphs, interaction data). The AsyncPostgresSaver checkpointer automatically persists GraphState snapshots to PostgreSQL after every GraphState update, indexed by a unique thread_id to identify each individual conversation thread [37]. This enables seamless multi-turn conversations, crash recovery and inter-agent result sharing (e.g. SNA metrics available to Response Agent).

Long-term memory extracts persistent farmer profiles, preferences and stable facts across conversations using Mem0 backed by PGVector embeddings. Semantic similarity search retrieves conceptually related memories without exact keyword matching, yielding 26% accuracy gains over full-context settings [38]. Before the response agent formulates an answer based on current findings from other agents, it uses mem0 to fetch memories that are relevant to the current context. Using the retrieved long-term context, the response agent would be able to formulate a more personalized response, remembering stored farmer preferences and previous key insights. After a response is formulated, the response agent would use mem0 to create new semantic memories based on the current full interaction of user query to agentic response [39]. This cycle keeps the long-term memory up to date with every conversational turn without missing any key insights.

V. VALIDATION AND PRELIMINARY RESULTS

CowNet-AI was implemented and tested on a local system with an 11th Gen Intel®Core TM i7-1165G7 processor (2.8 GHz, 4 cores, 8 threads) and 16 GB of RAM. All agents used the OpenAI gpt-4o-mini [40] for their LLMs. The simulation engine was first run for multi-week horizons to verify that emergent activity patterns and interaction rates were biologically plausible. The distribution of time spent in resting, feeding, moving, socializing, and drinking states was

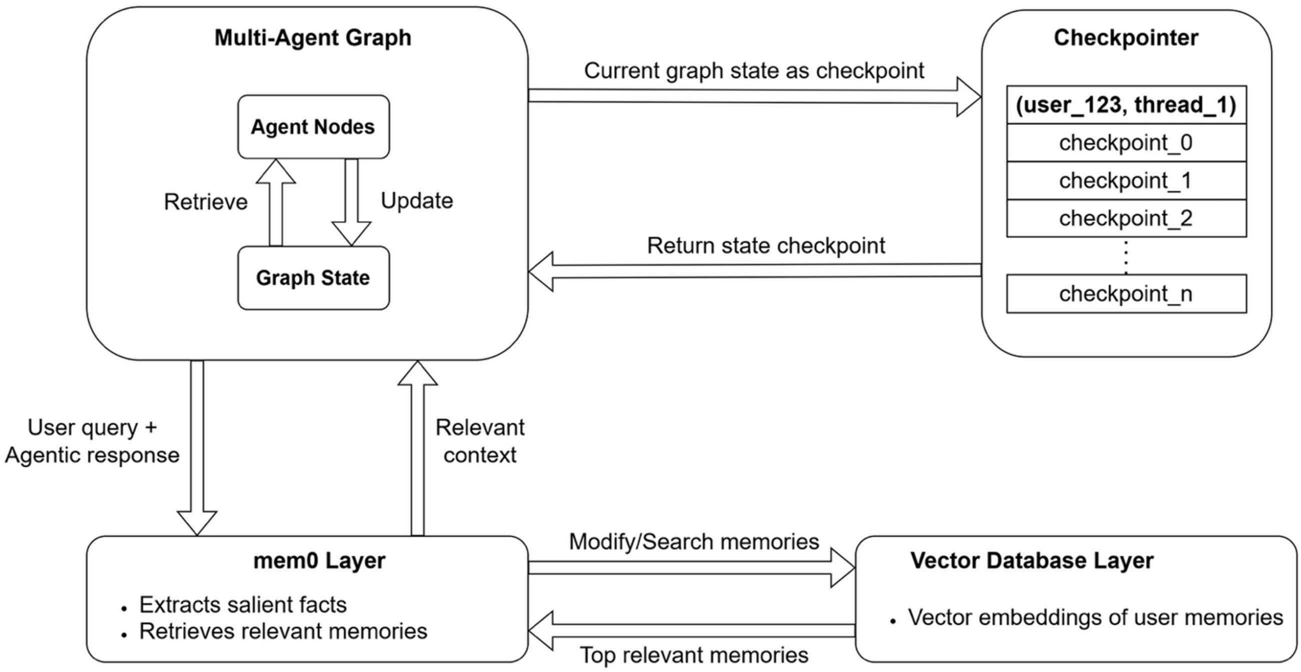

FIGURE 5. Graph memory and state management pipeline showing agent access to shared execution state, checkpointing, and long-term memory retrieval via vectorized storage.

consistent with published observations of freestall herds. Degree centrality values spanned a realistic range, indicating natural variability in social engagement. CowNet-AI is designed to generate the social network for the dairy herd using their interactions, computing the SNA metrics to specify their social engagements, and generate appropriate recommendations from these calculations and analysis. Hence, all inferences are based solely on SNA metrics derived from proximity logs, with no separate raw feature or temporal-smoothing model.

The SNA pipeline was applied to the MmCows dataset [41], which contains ultra-wideband location traces for 16 cows over 14 days. The sensor dataset was used for the CowNet-AI experiments using the 3D locations of the cows with the timestamps. The distances between every pair of cows were computed from their location data and the durations were calculated from the timestamps. For each pair of cows, an interaction was flagged if their distance was within 150 cm for $\geq$60 consecutive seconds. The interactions for each cow (with cow ID T01, T02, etc.) were then applied for further SNA analysis. Although the dataset includes 16 cows in total, due to consistent location data unavailability for 6 cows, 10 cows were considered for the complete experiments and analysis. Accordingly, these results should be interpreted as a pilot-scale validation rather than a statistically generalizable population study.

A. SNA AND ISOLATION RISKS

The social network analysis centrality measures were applied to the MmCows dataset by dividing the 14 days dataset into "Week 1" and "Week 2" data. This division enabled the visualization of the network changes between two consecutive weeks. As our interaction dataset showed that all cows were connected to each other (with different interaction frequencies), we removed one cow (Cow ID- T06) only for visualization of the network connections assuming one cow was removed from the pen in Week 2. Figure 6 shows the social network (undirected) for 10 cows from week 1 and week 2 (after removing cow T06 from week 1 network) using their degree centrality.

Then we used the previously divided week wise data (first 7 days and last 7 days) for centrality analysis (Table 2). The degree centrality, betweenness centrality, and the eigen vector centrality were computed for each cow separately for week 1 and week 2. The Δ values for each centrality was measured to show the differences between the week 1 interaction network and week 2 interaction network. Finally, the isolation score was calculated from the closeness scores showing the isolation risks in week 2. The resulting centrality measures and community structures were examined visually and against farm notes. The degree centralities from both weeks show the aforementioned property of a fully connected network.

B. QUERY RESPONSE

Finally, we uploaded the MmCows data to the CowNet-AI and provided the system a set of queries to check the responses of our multi-agent system. A sample input-output pair is shown in Figure 7 retrieved from LangSmith chat studio. When asked to show the top 3 most isolated cows,

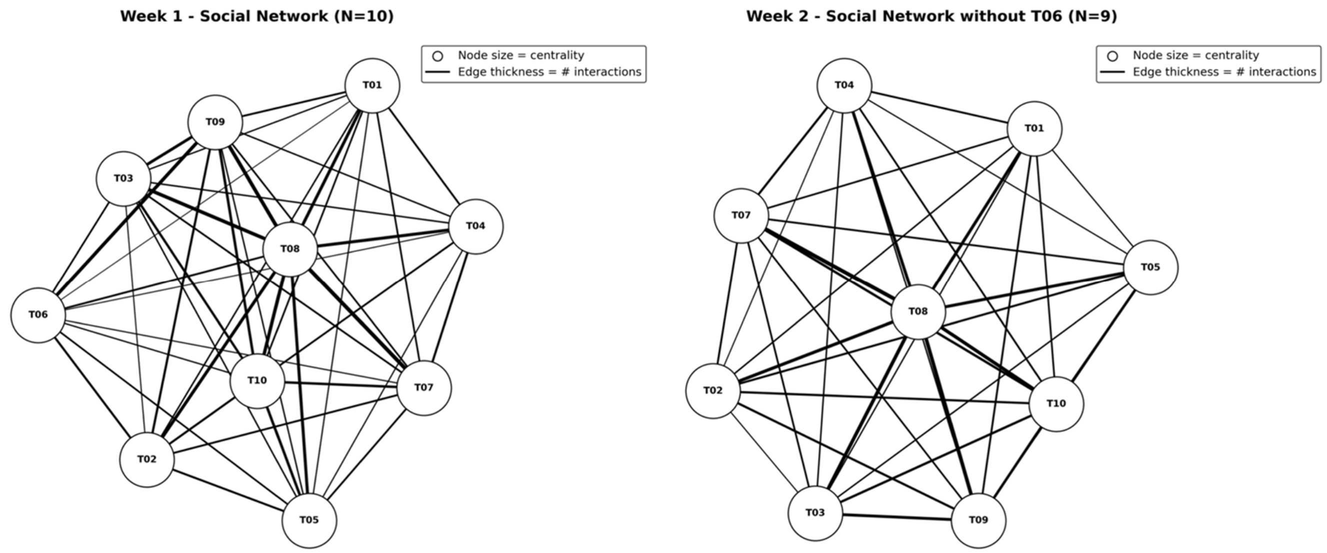

FIGURE 6. Social interaction network for Week 1 (N = 10) and week 2 (N =9, after removing cow T06. Node size represents centrality and edge thickness indicates interaction frequency.

TABLE 2. Centrality metrics (MmCows): Degree, betweenness, and eigenvector centrality measures for each cow in week 1 and week 2, the Δ values for each cow after week2, and the isolation score.

Cow ID	Degree W1	Degree W2	Δ Degree	Between W1	Between W2	Δ Between	Eigen W1	Eigen W2	Δ Eigen	Isolation Score
T01	1.0	1.0	0.0	0.111	0.056	-0.056	0.233	0.293	0.060	0.479
T02	1.0	1.0	0.0	0.0	0.0	0.0	0.326	0.257	-0.069	0.519
T03	1.0	1.0	0.0	0.0	0.139	0.139	0.356	0.237	-0.118	0.546
T04	1.0	1.0	0.0	0.139	0.042	-0.097	0.247	0.286	0.039	0.493
T05	1.0	1.0	0.0	0.028	0.0	-0.028	0.260	0.304	0.045	0.459
T06	1.0	1.0	0.0	0.250	0.111	-0.139	0.242	0.270	0.029	0.494
T07	1.0	1.0	0.0	0.0	0.0	0.0	0.276	0.355	0.078	0.497
T08	1.0	1.0	0.0	0.0	0.0	0.0	0.457	0.428	-0.028	0.571
T09	1.0	1.0	0.0	0.0	0.0	0.0	0.362	0.333	-0.028	0.483
T10	1.0	1.0	0.0	0.0	0.0	0.0	0.333	0.352	0.019	0.497
	Mean Δ Degree = 0.0			**Mean Δ Between = -0.0181**			**Mean Δ Eigen = 0.0027**			**Highest Isolation Score = 0.571**

the supervisor correctly routed to the data loader agent to load interactions into GraphState. After that the supervisor routed to the SNA agent to pre-compute SNA metrics for persistent use downstream. The supervisor decided that research, simulation and report agents were not needed, so it routed to the response agent, which used the query and SNA metrics from GraphState as context to devise an organized response back to the user.

As shown in Figure 7, the agentic, explainable AI - based CowNetAI generates explainable outputs from the computed SNA scores using the herd interaction data. The LLM translates these quantitative SNA metrics into concise textual explanations and potential animal welfare risk-based recommendations. The LLM-based interpretations and implications are not randomly generated. The example query and response show that the system generated responses are well established behavioral interpretations grounded in observable data and they provide clear and understandable explanations to farmers.

In this work, explainability is defined at the system and decision-support level, rather than as post-hoc feature attribution within a predictive model. CowNet-AI provides transparency by explicitly exposing the behavioral metrics, graph structures, and rule-based logic underlying each recommendation, and by generating human-readable explanations grounded in these computed quantities. This design emphasizes traceability and interpretability of decisions over model-internal attribution techniques.

C. EVALUATION PROTOCOL

CowNet-AI performance was evaluated (Table 3) using a comprehensive protocol that systematically compares the

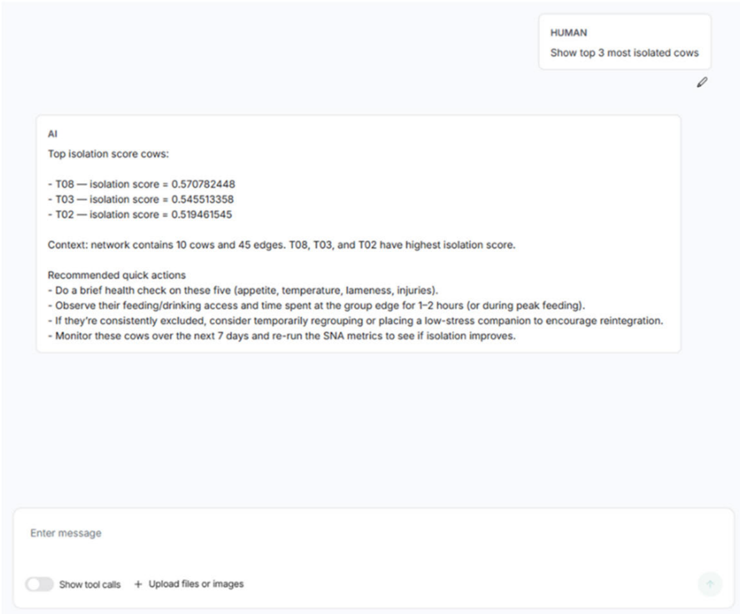

FIGURE 7. A sample input query and output response from CowNet-AI from the LangSmith chat visualizer.

multi-agent system against two systems: a single-agent LLM (GPT-4) [42] that receives the full SNA snapshot in a system prompt and generates responses in one shot without agent orchestration, memory persistence, or specialist computation; and a static rule-based SNA system that uses fixed keyword rules to route queries to predefined intents without LLM reasoning or dynamic state management, relying solely on the previously mentioned fixed intents and pre-computed SNA metrics. Experiments used query sets constructed via Perplexity AI Pro [43] based on fixed intents proportionally distributed across five categories: (i) isolation risk analysis (∼33%), (ii) centrality analysis (∼33%), (iii) what-if scenarios (∼17%), (iv) general explanations/herd management advice (∼10%), and (v) summary report generation (∼7%).

Evaluation metrics leveraged LLM-as-a-judge methodology, where an LLM (Perplexity AI Pro) acts as an impartial assessor (Table 3). This LLM received consistent system descriptions, persistent SNA metrics from MmCows week 2 SNA graph, and query-response pairs to score performance objectively across systems. Task success rates measured complete, justified responses (1=success, 0=failure); intent routing accuracy verified query-response alignment (1=match, 0=mismatch); hallucination rates flagged deviations from ground-truth metrics (1=hallucination present, 0=grounded); and latency captured full execution traces via LangSmith (multi-agent) or timestamp deltas (baselines). This enables rigorous ablation studies and head-to-head comparisons [27], [44]. The evaluation workflow follows principles proposed in global agent-governance frameworks, emphasizing task success, tool-call reliability, and context-driven assessments [28]. While LLM-as-judge evaluation introduces potential bias, it is used here as a preliminary comparative assessment framework, and we mitigate this risk by (i) providing identical system context and ground-truth SNA metrics to the evaluator across all systems, and (ii) restricting evaluation to factual consistency, intent alignment, and justification completeness rather than stylistic quality.

D. ABLATION STUDY

The multi-agent CowNet-AI system was evaluated using an ablation study by computing the performances for the complete system, system without the supervisor agent, system without the research agent, system without the simulator agent, a direct single agent LLM (GPT-4), and a rule-based static SNA (Figure 8). A total of 20 queries were used, with the comparison being made over the overall task success for each configuration.

Results show that the complete system and the system without the supervisor achieved 100% task success. This shows the importance of each CowNet-AI specialized agent as these are the only 2 configurations to include all specialized agents. The full system has a supervisor that can route and access any specialized agent, and without the supervisor, the system would be configured to route through all specialized agents sequentially regardless of whether that agent is needed. The distinction with the supervisor addition suggests that the key advantage lies in efficient routing and latency optimization, by ignoring redundant routing to unused agents.

Without the research agent, we see task success decrease to 60%. A condition for counting a task success is the inclusion of justification of the response, in addition to the correctness. Given the research agent's role in extracting supportive facts and evidence, its absence would suggest

TABLE 3. Summary of evaluation protocol.

Evaluation Component	Definition	Implementation
Query Construction	Process of developing test queries	LLM-generated queries on- • Isolation risk analysis, • Centrality analysis, • What-if scenarios, • General explanations/herd management advice, • Summary report generation.
Task Success Rate	Response rates for correct, justified answers	LLM-as-judge • 1=success, • 0=failure.
Intent Routing Accuracy	Rate of correct intent classifications	LLM-as-judge verifies query-response intent match • 1=match, • 0=mismatch.
Expert-rated Relevancy	Average Likert (1-5) score for relevance or explainability	LLM-as-judge trustworthiness scoring
Hallucination Rate	Rate of responses with misinformation or lack of justification	LLM-as-judge vs. SNA metrics context • 1=hallucination, • 0=grounded.
Latency	End-to-end response time	• LangSmith traces (multi-agent) • Timestamp deltas (baselines)

a lack in evidence-grounded responses, affecting the task success score.

Without the simulation agent, task success rate is 85%. As this agent's only role is to support "What-if?" scenario-based intents, it would be expected that the system would not respond well to those queries without their specialized agent. All other intents would be handled by all other agents, a reasonable assumption for this configuration.

The single-agent LLM (GPT-4) achieved 50% task success rate, just slightly less than the CowNet-AI configuration without the research agent. This is explained by the lack of tools a single-agent LLM has. A noticeable missing feature is a simulation tool, which the configuration without research agent had, allowing it to justify "What-if" scenario queries with simulation-modified SNA graph metrics.

Lastly, the rule-based static SNA configuration achieved the lowest task success rate of 35%. The main factors affecting this result are the lack of tool access to perform tasks like open-ended research or simulations, and the lack of natural language processing, relying on keyword-matching with no semantic inference of query meaning. This massively restricts this system's performance as it is limited to a finite set of keywords that would invoke a response. However, if an intent is matched by a keyword, pre-defined explanations and responses, alongside pre-computed SNA metrics would result in the success of some queries.

E. AGENT AND MODEL PERFORMANCE COMPARISONS

The full CowNet-AI system was also tested locally to compute and evaluate the performance of each agent separately (Figure 9). Total 100 queries were used to evaluate and compute the average latency breakdown. Figure 9. reveals characteristic execution profiles across the CowNet-AI workflow. The research agent exhibits the highest average latency (80.9 ± 54.0 s) due to iterative external API calls to Semantic Scholar, arXiv, and PubMed, retrieving voluminous metadata payloads (full abstracts, citations) with potential re-querying for sufficient coverage. Conversely, data loader agent (0.11 ± 0.04 s) and SNA agent (0.06 ± 0.04 s) demonstrate minimal latencies as they execute exactly once per conversation—loading fixed-schema interactions from local-directory location data and computing comprehensive NetworkX metrics that persist via GraphState for downstream reuse, eliminating redundant computation.

Report agent (24.0 ± 5.1 s) reflects PDF assembly overhead from aggregating SNA metrics, research summaries, and visualizations. Response agent (20.6 ± 7.2 s) incurs synthesis time for farmer-friendly explanations drawing from full GraphState context. Simulation agent (10.7 ± 8.6 s) variability stems from graph manipulation complexity (cow removal scenarios). Supervisor (7.5 ± 3.6 s) balances LLM reasoning across intent inference and state inspection. These profiles validate the GraphState persistence design, minimizing redundant computation while isolating external API variability to research-specific workflows.

The model was also compared against the rule-based static SNA and the single agent LLM (GPT-4) on the same subset of 20 queries used in section D (Table 4). The results reported in the table are mean values over 3 runs, along with 95% confidence intervals. CowNet-AI significantly outperforms the generative single-agent LLM, as well as the rule-based static SNA across key dimensions,

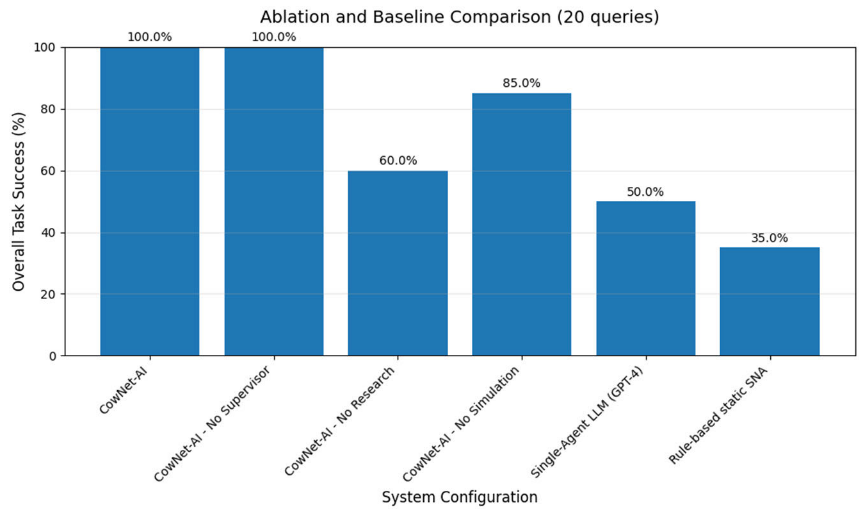

FIGURE 8. **Results of ablation experiments where individual agents are disabled to quantify their contribution to overall system performance. The system performance is shown for the full System (all agents enabled), No Supervisor Agent, No Research Agent, No Simulation Agent, Single-agent LLM (GPT-4), and Rule-based static SNA.**

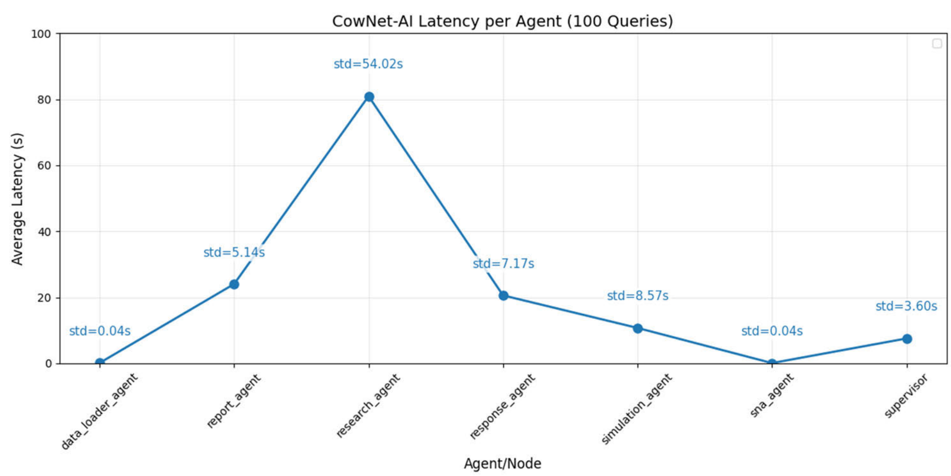

FIGURE 9. **Average execution latency per agent measured over 100 queries. Near-zero latency reflects cached or precomputed operations, while higher latency highlights computationally intensive agents.**

demonstrating the value of multi-agent orchestration for production-grade dairy SNA (Table 4). With 100% intent routing accuracy and 5/5 expert-rated relevancy, CowNet-AI achieves perfect task understanding and farmer utility, compared to 80% and 3/5 for single-agent GPT-4, and 65% and 2/5 for static rules. The 5% hallucination rate-near ground-truth fidelity—contrasts sharply with GPT-4's 40% hallucinations, validating GraphState persistence and specialist agent grounding against LLM free-form generation. The hallucination rate in CowNet-AI is not relevant to the SNA computations and solely depends on the lack of justification or misinformation provided by the LLM while generating text-based responses for the user. The deterministic nature of the static rule-based SNA system results in no hallucination as responses and intents are pre-determined with no semantic reasoning or natural language processing.

Latency trade-offs reflect architectural priorities. CowNet-AI's 58s average stems from research agent API calls (80.9s) and comprehensive workflows yet remains practical for weekly farm briefings. Static rules deliver <1s via template matching but sacrifice adaptability (35% routing failures). GPT-4's 7s balances speed and flexibility but fails on factual consistency, underscoring multi-agent benefits for high-stakes agricultural decision support where accuracy trumps marginal latency.

These demonstrations confirm that the agents can collaborate to answer open-ended management questions, and that the explanations produced are grounded in computed metrics.

TABLE 4. System comparison for CowNet-AI vs. baselines (rule-based static SNA, single agent LLM) with their average routing accuracy, average explanation rating, average hallucination rate, and average query latency (in seconds) for 20 queries based on MmCows dataset.

Model	Average Intent Routing Accuracy	Average Expert-Rated Relevancy	Average Hallucination Rate	Average Query Latency
CowNet-AI	100%	5	5%	58s
Rule-based static SNA	65%	2	0%	<1s
Single-Agent LLM (GPT-4)	80%	3	40%	7s

Performance metrics reported in Table 4 are computed over a fixed evaluation set of 20 queries derived from predefined intent categories. For binary outcomes (e.g., task success, hallucination presence, intent routing accuracy), 95% confidence intervals (CI) were estimated using the Wilson score interval for binomial proportions [44]. For example, the observed hallucination rate of 5% (1 out of 20 queries) corresponds to a 95% CI of approximately 0.9%–23.6%, reflecting sampling uncertainty associated with the limited evaluation size. Similarly, a 100% intent routing accuracy (20/20) corresponds to a 95% CI lower bound of approximately 83.2%. These intervals indicate that while point estimates are strong, larger-scale evaluations would further refine precision estimates. Latency statistics are reported as mean ± standard deviation across 100 independent query executions to capture runtime variability.

F. SCOPES AND LIMITATIONS OF WELFARE INFERENCE

CowNet-AI does not perform supervised classification of animal welfare states and does not rely on veterinarian-annotated labels, physiological biomarkers, milk yield records, cortisol measurements, or other direct clinical indicators. Instead, the system derives structural social network metrics, including degree deficit, centrality decline, closeness deficit, and interaction frequency deviation, to quantify changes in herd interaction patterns. The resulting isolation risk score should be interpreted as an explainable, behaviour-based early-warning indicator rather than a clinically validated diagnostic measure. It represents a graph-structural assessment of social vulnerability and deviation over time, rather than a direct measure of welfare status. The system is designed to flag statistically significant changes in social positioning that may warrant targeted inspection or follow-up by farm personnel. Accordingly, all recommendations are advisory and intended to support, not replace, farmer and veterinarian judgement. No claim is made that the isolation risk score constitutes a ground-truth welfare label. Future work will focus on integrating expert annotations, health records, and longitudinal outcomes to establish quantitative links between social network perturbations and clinically validated welfare states.

VI. DISCUSSIONS

CowNet-AI illustrates how a multi-agent LLM architecture can be combined with social network analysis to create an explainable decision-support system supporting farmers in identifying potential dairy cattle welfare risks. By integrating sensor-based data, graph-based behavioral metrics, simulation capabilities, and research grounding, the system helps move from descriptive analytics toward proactive management.

The proposed system framework enhances authenticity and response retrieval robustness through a controlled multi-agent design that separates data access, SNA and contextual response text generation. This task decomposition ensures that all responses are based on farm data and the computed SNA metrics. The supervisor agent enforces relevance validation and robust query routing among other agents to reduce hallucinated outputs.

From a scientific perspective, the platform supports new types of analysis of herd social structure and its relationship to health and productivity. From a practical perspective, CowNet-AI offers a way to embed complex analytics into a conversational interface that aligns with how farmers and consultants naturally ask questions. Simulation-based "what if" analysis allows users to explore the likely consequences of group changes before implementing them, which may reduce costly trial-and-error.

At the same time, several technical and ethical challenges remain. Large language models can hallucinate, and even with careful prompt design and data grounding, there is a residual risk that the system may suggest actions that are not supported by evidence. We therefore explicitly design CowNet-AI as an advisory system: farmers and veterinarians retain ultimate authority, and recommendations are presented with context and uncertainty. Although the latency for different agents is justified by the need for accurate results in a sequential system design, the use of high-performance hardware, incremental SNA graph updates, recomputation of only required metrics, as well as asynchronous agent execution and local knowledge caching strategies, can reduce overall query execution time. Computational cost is another challenge, although preliminary experiments with GPU acceleration are promising. Data quality, labelling limitations, and representativeness across different housing systems must also be addressed to ensure that computed metrics are robust and generalizable.

Additionally, future CowNet-AI deployment in dairy farms will include storing and managing larger animal data and cumulating conversations over time. The simple centralized shared GraphState architecture simplifies the agent coordination and faster responses in the pilot study and include basic mitigation for bloat by each agent using only the relevant state variables needed for the specific task. However, it may introduce state bloat due to the accumulation of frequent farmer interactions. Production deployment can incorporate memory-control features to enhance scalability of the system by including specific sliding-window truncation of the user conversation history, aggregating raw interactions logs into time-based summaries, and storing all raw data, computations, and conversation history into external databases with references. Furthermore, for larger herds, scalability enhancement is needed to manage computational complexity. The complete computation complexity for network generation and SNA metrics computation for all pairwise interactions can reach to $O(N^2)$ for herd size N. Hence, including incremental graph updates, sparse network representations, potential computation parallelization, and GPU-accelerated SNA calculation can help maintaining optimal user responses.

The 5% residual hallucination rate, while low compared to the other two baseline (generative single-agent LLM and pre-defined rule-based static SNA) systems (40%), remains non-zero for a safety-critical domain. CowNet-AI addresses this through: (1) explicit design as an advisory system where farmers retain ultimate authority, (2) recommendations presented with uncertainty context and supporting evidence, (3) persistent memory that tracks recommendation acceptance/rejection to improve calibration, (4) integration of high performance reasoning model or LLM, and (5) additional constraints for structured response generation. Production deployment would require additional human-in-the-loop verification protocols.

While CowNet-AI provides decision-level transparency through explicit metric reporting and traceable agent orchestration, it does not currently implement formal model-internal explainability techniques such as SHAP, LIME, or counterfactual attribution. These methods are most applicable to black-box predictive models, whereas CowNet-AI's primary decision logic is derived from symbolic social network metrics and rule-based reasoning. Integrating formal attribution and uncertainty calibration mechanisms into the agentic pipeline, particularly for hybrid predictive components, is an important direction for future work.

Finally, this study utilizes a publicly available benchmark dataset (MmCows) and does not include any new experiments on dairy cattle. However, any future deployments, reproductions and/or improvements of CowNet-AI involving custom datasets should follow animal welfare regulations, data governance policies, and ethical approval requirements. Responsible, transparent, and ethical animal welfare monitoring practices are essential for practical deployments.

VII. CONCLUSION AND FUTURE WORKS

CowNet-AI advances precision dairy farming by uniting social network analysis, multi-agent large language models, and explainable decision support in a single platform. CowNet-AI is built to work alongside the farmers to monitor cow welfare risks by analyzing their social behaviors and interactions to provide timely alerts and recommendations. It is designed to assist humans, and not to replace them in any way, ensuring that the farmer remains the final decision-maker. The system represents, to our knowledge, the first agentic architecture for livestock social network analysis that is both open-source and explicitly designed for transparency. By supporting both fixed-intent and open-ended queries, and by grounding recommendations in metrics and literature, CowNet-AI provides a practical route for incorporating complex analytics into farm management. Outperforming standard single-agent LLM like GPT4 and basic rule-based static SNA models in system-level evaluations by 20% to 35% accuracy in query responses, CowNet-AI shows promising performance for real-time farm applications while processing each query in less than a minute. The agent-specific tests show the robustness of a multi-agent system for providing dependable responses.

Future work will focus on integrating automatic behavior recognition to better differentiate positive and negative interactions, extending to multimodal data sources such as vocalizations and health records, and exploring privacy-preserving training regimes such as federated learning. Potential future efforts will also include alternative deployment architectures (cloud vs. local/edge); LLM backend substitutions; advanced and user-friendly UI; and streaming and scaling considerations built upon the proposed and tested baseline system. We also plan to map outputs onto formal welfare assessment frameworks to support regulatory and audit processes. To support robust real-world deployment, future will also incorporate standardized protocols and evaluations for sensor calibrations ensuring data consistency and reliability. Advanced methods for data imputation, noise-filtering, and outlier handling approaches will be integrated to enhance the robustness of cow localization, movement tracking, and the SNA metrics. Furthermore, large-scale, multi-site, longer studies in different seasons with various cow breeds, herd compositions, and environmental conditions will be conducted to validate the reproducibility and generalizability of CowNet-AI. Through these extensions, we aim to demonstrate that agentic, explainable AI can deliver measurable improvements in dairy cattle welfare assessment and in the quality of farm decision making.

DATA AND CODE AVAILABILITY

The source code supporting this work is publicly available at https://github.com/mooanalytica/COWNET-AI-MultiAgent under an open-source license. The MmCows dataset used for validation is publicly accessible at https://github.com/neis-lab/mmcows

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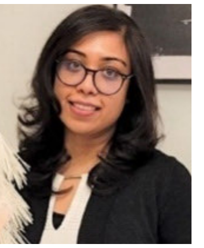

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